Trace Resistance and Blowing Up Traces

ECEN 3730 - Henry Matar

2 vs 4-Wire Measurement

- Two-wire setup measures the resistance across the trace
- Four-wire setup induces current through the trace and determines the voltage drop to calculating resistance

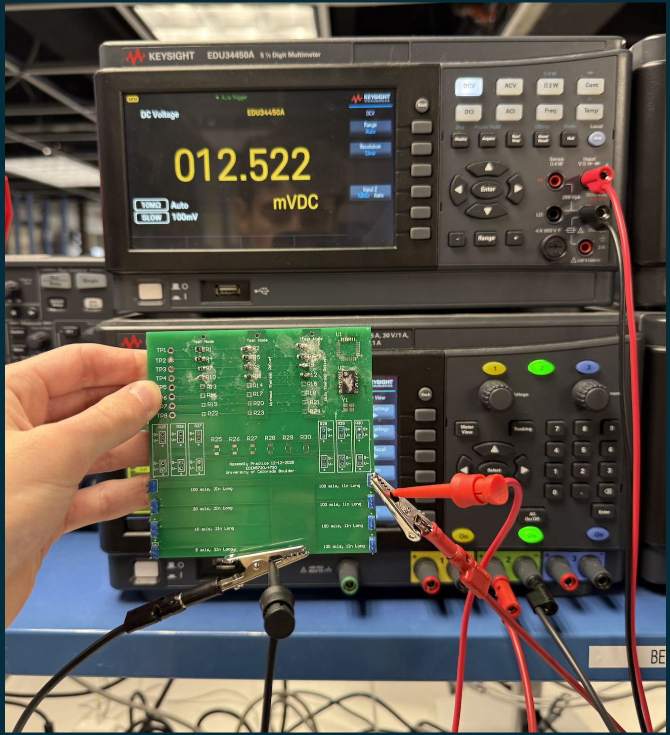

Fig. 1 Measurements using Four-Wire Setup

Measured Results

Line Length	Line Width	Estimated Res	2-Wire Measurements w/o Probe Nulled	2-Wire Measurement w/ Probe Nulled	4-Wire Method (R=V/I) Voltage (mV)	Current	Resistance
1 Inch	5 mil	0.1	0.375	0.26	55.03	0.999	0.05508508509
	10 mil	0.05	0.156	0.064	31.86	0.999	0.03189189189
	20 mil	0.025	0.0201	0.108	30.9	0.999	0.03093093093
	100 mil	0.005	0.183	0.086	14.25	0.999	0.01426426426
2 Inch	5 mil	0.2	0.313	0.22	223.1	0.999	0.2233233233
	10 mil	0.1	0.209	0.116	107.4	0.999	0.1075075075
	20 mil	0.05	0.16	0.067	56.19	0.999	0.05624624625
	100 mil	0.01	0.137	0.045	18.62	0.999	0.01863863864

Measurements Explained

- Rule of thumb provided a baseline to utilize rule #9 for all the measurements
- Baseline two-wire measurement was the least accurate as it doesn't account for resistance in the probe
- The nulling in the two-wire measurement results in increased accuracy, removing error
- The four-wire measurements yielded the most accurate results by inducing a current and measuring resistance from that
- Why Copper Thickness contributes to uncertainties
  - Copper thickness controls the amount of current a trace can carry, affecting resistance based on thickness
  - When dealing with small traces, minute inconsistencies can affect resistance substantially causing discrepancies between our measurements and calculations

Burning Traces

- To burn the trace a current of 4A and 5V was induced using the desktop power supply
- Using the special clamps capable of withstanding the large current I applied it across the trace and waited until it burned both visually and through smell
- The trace heated up quickly at a lower current and when current increased it burned in a matter of seconds

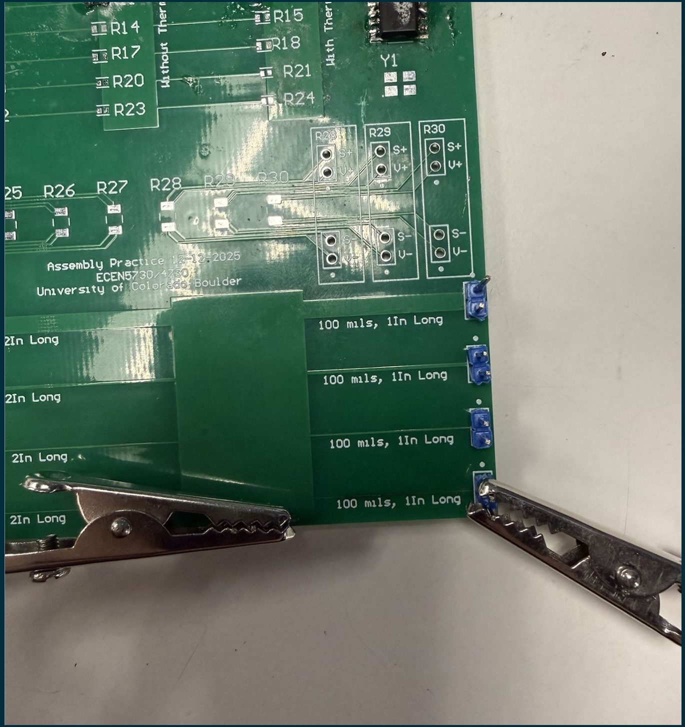

Fig. 2 Burned Trace

Trace Fatigue & Saturn Estimates

- Saturn gave an estimate of the 5mils trace burning at ~865 mA
- I applied a current starting there and incrementally increasing until the trace burned
- At 4A the trace burned in 27 seconds which is far greater than Saturn's conservative estimate
- The trace changed state as current increased, getting hot then eventually burning.
- I recommend a maximum of 2A through the trace

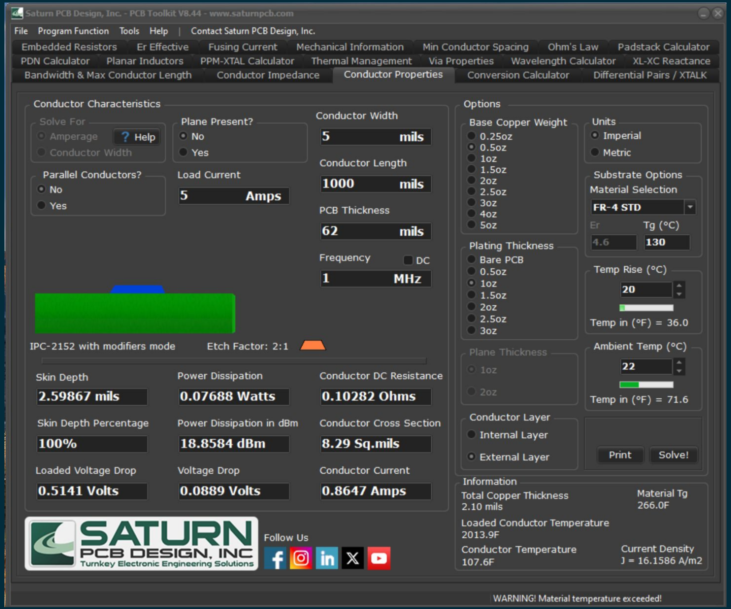

Fig. 3 Saturn Tool Estimates

Summary

- Two-Wire measurement is common as it is easy to setup and sell in DMM
- This setup is still useful in most applications
- Four-Wire is the most accurate setup but it more complex to setup and harder to sell commercially
- Saturn estimate was extremely conservative (865 mA)
- The trace blew up at 4A and an estimated max current would be ~2A